IoT based Soil Monitoring & Irrigation System for Agriculture

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Abstract—Agriculture is the main resource of the survival of India. This research proposed IoT based smart agriculture systems which help to increase crop production. In addition, Internet of Things (IoT) enabled automated irrigation systems aimed at optimizing water utilization and overcome from the wastage of the water and simultaneously protecting crop health through continuous soil moisture level and requirement of the water in specific area. IoT based systems incorporate soil moisture sensors, microcontrollers, and advanced data processing procedures to enable the automated supply of water. The received sensor data and by using automated decision-making processes, this system improves irrigation efficiency as well lower the labor cost. It will also help in improving farmers' incomes through increased crop production. Implementation of remote sensing technologies along with real-time data analysis enables truthful water distribution. Also, based on the results received by the sensors data farmers can cultivate the crop which suits the specific soil.

Keywords: Soil Monitoring, Temperature Sensor, Humidity Sensor, Crop Images, Sensor Measurements, Soil Test.

I. INTRODUCTION

Agriculture is the economic backbone of economies worldwide, providing food security and livelihood to billions of people. However, agriculture sector has numerous critical problems, one of which is the efficient utilization of water resources. Traditional irrigation systems based on manual control and fixed scheduling tend to lead to inefficient use of water, resulting in over-irrigation or under irrigation. Over irrigation leads to waterlogging, filtering of nutrients, and increased sensitivity to plant diseases. While under-irrigation leads to plant stress, low yield, and poor crop quality. Such inadequacies not only cause waste of water but also restrain agricultural productivity, especially in regions already miserable from water shortage due to climate change, population growth, and urbanization at a rapid rate etc. [1-3].

Water scarcity is a serious global issue, and agriculture applies approximately 70% of the freshwater available in the world [5]. Incompetent irrigation systems intensify the issue, and therefore smarter & viable water usage systems need to be used. Farmers typically employ experience or traditional practices to set irrigation schedules, which may vary from the actual moisture needs of the soil. This lack of precision in water relevance not only puts excessive stress on water resources but also impacts crop health and yields [1].

To overcome such issues, the application of emerging technologies such as the Internet of Things (IoT) in agriculture has been a game-changing solution. IoT-based smart irrigation systems use real-time information from soil moisture sensors, weather, and microcontrollers to optimize and automate irrigation [2]. The proposed system supplies the required water to the crops as per requirement and minimizes water loss and maximizes process efficiency. Since IoT based systems constantly monitor soil moisture and environmental conditions, IoT-based systems can trigger irrigation whenever it is necessary, in that way formers able to save water and improve crop health [3-4].

The main components of an IoT-driven smart irrigation system include soil moisture sensors, microcontrollers (e.g. ESP32 or Arduino), water pumps, and electrical relay modules. Such devices are used to create an independent system that operates with marginal human control. These systems can also be connected to cloud platforms and smartphone apps so that the farmers have remote access for monitoring and regulating irrigation operations, enhancing access convenience and efficiency. The proposed system can be connected to predictive analytics and weather data to enhance irrigation schedules depending on prevalent weather conditions [2] [6].

There are lots of benefits of IoT-based smart irrigation systems. Studies have shown that they can conserve water by as much as 40% and increase crop yields by 20–30% [1]. Also, they lower the cost of labor and eliminate human errors, thus making agriculture more cost-efficient and sustainable. Moreover, utilizing renewable energy sources, such as solar panels, makes such systems operational even in large areas with minimum access to electricity [4].

Though the proposed system has numerous benefits but the large-scale adoption of IoT-based irrigation is being hindered by a variety of challenges. The challenges include high initial cost, reliance on strong internet connectivity, and a lack of technical expertise among farmers etc. It will be crucial to achieve the proposed systems goals of smart irrigation technologies to address such hindrances through cost-effective solutions, hybrid connectivity, and farmer training schemes [3] [7].

This research work delves into the design, development, and benefits of an IoT-based smart irrigation system. Trying to overcome the limitations of traditional irrigation systems and tapping the potential of IoT. The proposed methods may help make farming more sustainable and efficient. The findings of the research show the potential of smart irrigation systems to conserve water, increase crop yield, and promote

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environmental sustainability, we can usher in a greener and smarter era of farming.

II. LITERATURE REVIEW

Agriculture is the backbone of India, but the main challenge is the water scarcity. In agriculture regular use of water generally is the cause of wasting water which affects both crops and soil. So, monitoring and management of water is the need of time to avoid the wastage of the water and hence to implement proper irrigation system needs to be proposed and implemented. The problem can be resolved by smart agriculture by using and IoT. Smart agriculture systems through a sensor will be able to compute required quantity of water and nutrients for supply into the growing area of the crops. The proposed methodology not only mitigates concerns related to water scarcity but also supports sustainable agricultural systems. [6] [8].

In the last few year, the application of Internet of Things (IoT) in agriculture has gained significant consideration with an emphasis on irrigation management. Researchers and experts have suggested the concept of utilizing IoT-based systems for the improvement of inefficiency of traditional irrigation methods based on manual control and fixed timers. The limitation of the existing traditional irrigation systems tends to over-irrigate or under-irrigate as a result of which water is lost, crop production decreases, and the environment is degraded [16-18].

Several studies [9] [19] have demonstrated the effectiveness of IoT based smart irrigation systems in optimizing water consumption and crop yield. An IoT-based smart irrigation system, which employed soil moisture sensors and microcontrollers to control irrigation. Their findings demonstrated that the system significantly reduced water consumption while maintaining optimal soil moisture levels, thereby improving crop health and yield.

There are some studies shown the significance of real-time monitoring of soil moisture in precision agriculture. Their work demonstrated that IoT-based systems had the ability to provide farmers with real-time information on soil conditions, which would allow them to make well-informed irrigation schedule decisions [6-7].

Over the several benefits there are challenges of IoT-based smart irrigation systems that constrain their acceptance. One of the significant barriers is the cost, high upfront cost of IoT devices, sensors, and cloud infrastructure, which is too expensive for small-scale farmers. In addition, other limitations are the stable internet connection and power supply especially in rural and remote areas. Power outages and network failures can weaken the performance of IoT systems, interpreting them ineffective maybe when there is the need [10-11].

To solve these problems, there are some solutions investigated by researchers and scientists. One of them is the use of solar powered sensors to provide continuous power to IoT-based irrigation systems where there is a chance of frequent power cut or supply of limited electricity. It can be an economic and renewable source of energy for IoT devices and thereby make them less expensive and more consistent to use. Another solution is to use open-source platforms such as Arduino and ESP8266 to make low-cost IoT-based irrigation systems [4] [10].

Another important aspect of IoT-based irrigation systems is that it can be monitored and controlled in real time through mobile apps and web portals. Apps such as Blynk enable farmers to remotely monitor soil conditions, receive alerts, and manually override irrigation schedules when it is necessary. This method is easy and convenient not only to reduce labor expenses but also eliminate human errors and hence make agriculture more economical and natural [13] [17].

In this paper author presented potential of IoT-based smart irrigation systems to solve the problems of usual irrigation systems. By analyzing real-time data and predictive analytics, with IoT and sensors, these systems are capable of optimum water competence and enhance crop yield. Even though high costs and implementation of the smart irrigation system are the issues that need to be overcome for the acceptance of smart IoT based agriculture.

III. DATA COLLECTION AND ANALYSIS

The proposed IoT-based soil monitoring and irrigation system collected environmental and soil condition data over a continuous time-period using embedded sensors. Key parameters logged included soil moisture (%), soil temperature (°C), pH level, and macronutrient concentrations (Nitrogen, Phosphorus, Potassium in mg/kg). Data were recorded at fixed time intervals and associated with the operational status of the irrigation system (on / off). Fig. 1 is presenting result of the implementation/

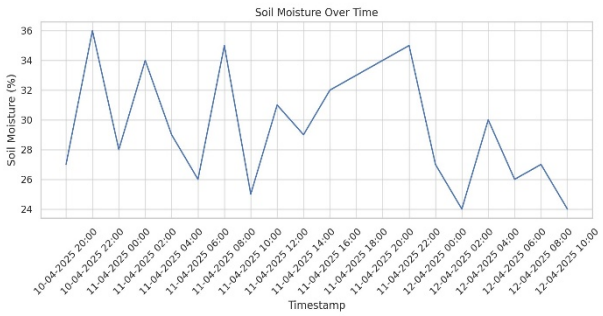


Fig. 1. Soil Moisture Over Time

Soil Moisture Over Time demonstrates the variation of soil moisture levels over time. It is evident that the system successfully identifies moisture thresholds, triggering irrigation when levels drop below the set limit. The immediate increase in moisture following irrigation events confirms the system's responsiveness and effectiveness [12] [15].

Fig. 2 showing nutrient analysis using a box plot. It displays the distribution and consistency of nitrogen, phosphorus, and potassium levels throughout the observation period. These

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insights are crucial for understanding soil fertility and planning fertilizer applications.

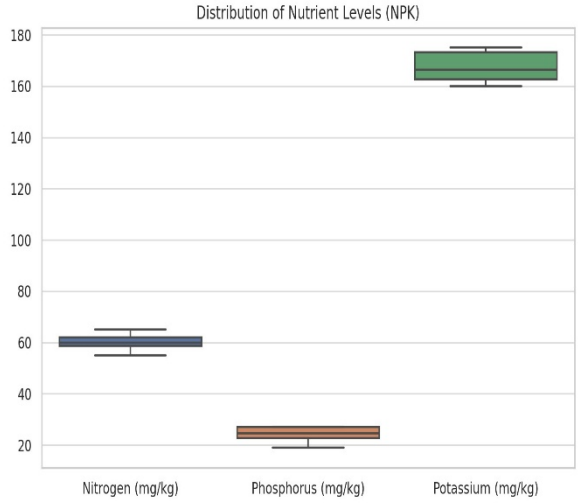


Fig. 2. Distribution of Nutrient Levels

IV. PROPOSED METHODOLOGY AND IMPLEMENTATION

The proposed design of soil monitoring and irrigation system executed an IoT-based smart irrigation system that maximizes water usage and increases the yield of the crops. The study is based on three key methodologies: Soil Moisture Measurement, Data Processing, and Automatic Irrigation. To design and implement this system the main components/devices needed like Soil Moisture Sensor, Soil Temperature Sensor, PH Sensor, NPK Sensor, Microcontroller, Relay Module, Water Pump, Power Supply, Breadboard, Connecting Wires etc.

Fig. 3 displayed the proposed workflow diagram of the whole embedded and IoT diagram. In this methodology data will be captured by the sensors and analyzed by the python coded web application. Based on data, analytics decisions will be taken about the soil condition. The water pump on/off control also activated on this methodology through web application. Once application is activated with related devices/components, then machine will be able to read all captured sensor data. Following steps involved in the proposed methodology-

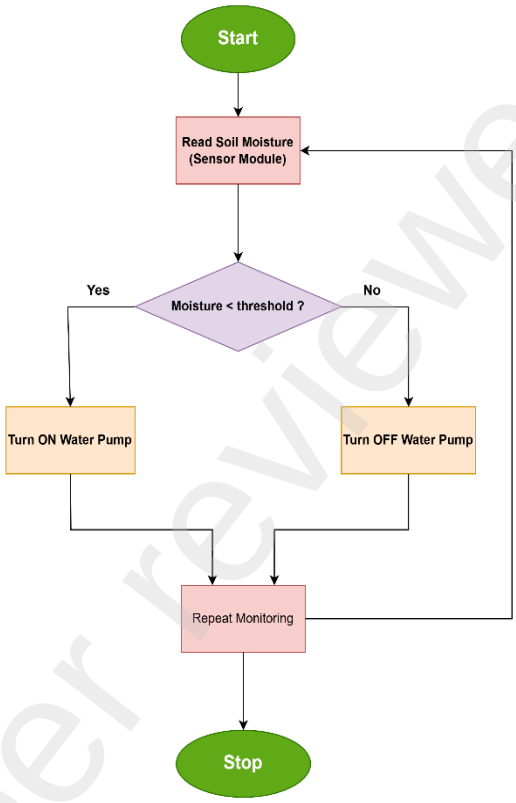


Fig. 3. System Architecture of the Working Model

A. Soil Moisture Measurement

The measurement of soil moisture is an essential component in any smart automated irrigation system. It is shown in the proposed architecture that how a comprehensive soil sensing mechanism has been employed, utilizing both capacitive soil moisture sensors and soil nutrient supervising sensors to improve the precision and adaptability of the irrigation strategy.

Since capacitive soil moisture sensors are durable, corrosion resistance and improved accuracy over resistive sensors therefore it is proposed system. These sensors detect the dielectric permittivity of the soil, which varies with moisture content. As per existing studies and research the sensor outputs an analog signal in the range of 0–1023, which is read via the analog input of the ESP32 microcontroller. The sensor values are interpreted in real-time, with low readings indicating dry soil and high readings indicating wet soil. These sensors are placed at a depth of 10 to 20 cm, which corresponds to the root zone of most cultivated crops. For heterogeneous soil fields, multiple sensors are deployed across various zones, and the average value is computed to ensure uniformity in irrigation decisions [8-10]. Fig. 4. representing the soil moister monitoring process.

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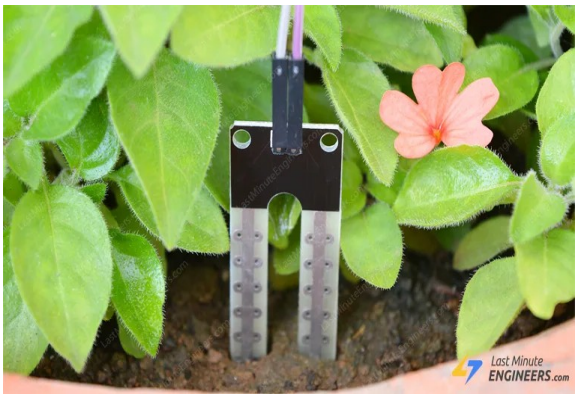


Fig. 4. Soil Moisture Monitoring [21]

Before deployment, each sensor undergoes a manual calibration process to establish its sensitivity curve under three key moisture conditions: dry soil, field capacity, and saturated soil. This process yields a calibration equation that converts analog values into estimated volumetric water content (% VWC). This allows the microcontroller to operate on more meaningful and consistent moisture values rather than unprocessed analog data [7] [12].

In addition to moisture sensing, the smart system integrates soil nutrient monitoring sensors particularly NPK (Nitrogen, Phosphorus, Potassium) sensors. These sensors can detect the concentration levels of essential nutrients directly from the soil. The sensors typically operate on electrochemical or optical principles, and output either analog or digital values corresponding to nutrient concentrations [14]. Integrating nutrient sensing allows for dual-purpose monitoring:

1) It informs the user about soil fertility in real time.
2) It enables the system to recommend or automate fertilizer application scheduling, thereby supporting precision agriculture.

The soil moisture and nutrient monitoring capabilities together provide a multi-dimensional insight into soil health, safeguarding that water is applied when needed and that the soil remains nutrient-rich for optimal crop growing. This dual sensor approach not only conserves water but also reduces unnecessary fertilizer usage, minimizes nutrient leaching, and promotes environmentally bearable farming.

B. Arduino-Based Data Processing

The Arduino microcontroller acts as the central processing unit for interpreting and managing the raw data obtained from the soil sensors. Data processing begins with the acquisition of analog signals from both the capacitive soil moisture sensors and the NPK nutrient sensors. These sensors output raw electrical signals that vary in voltage depending on the moisture level or nutrient concentration in the soil. Arduino reads these inputs as digital values ranging from 0 to 1023. To ensure consistent data sampling without interrupting system performance, nonblocking timing methods such as millis () are used to collect readings in regular intervals.

Raw sensor data is not inherently meaningful in its original form, so a conversion process is essential to translate these readings into understandable soil parameters. For moisture data, a calibration equation derived from pre-deployment tests is used. This equation maps the analog values to estimated Volumetric Water Content (VWC) percentages, allowing the system to report real-time soil moisture as a function of actual water availability. Similarly, readings from the NPK sensor are converted into nutrient concentration levels typically expressed in parts per million (ppm) or milligrams per kilogram (mg/kg), using manufacturer-provided scaling data or custom calibration formulas. This stage ensures that both moisture and nutrient data are accurate, consistent, and suitable to get analytics [14-17]. Fig. 5 represents the Arduino based data processing and analysis procedure system.

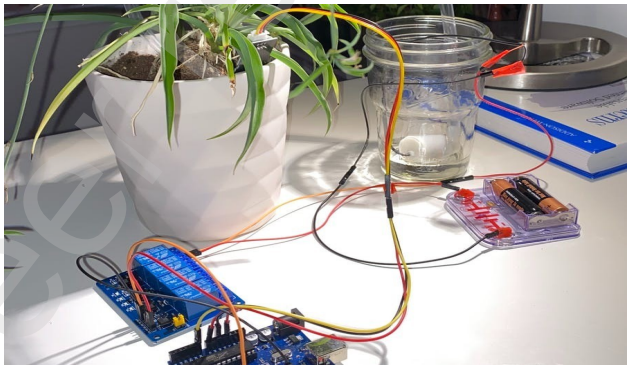


Fig. 5. Arduino-Based Data Processing and analytics [20]

Later stage, the Arduino compares the processed values against predefined threshold limits that are based on typical requirements of agricultural crops. For instance, a soil moisture level below 20% VWC might be classified as dry and trigger an irrigation response, while low nutrient concentrations could prompt a recommendation for fertilizer application. These thresholds can be customized based on crop type, soil conditions, or user preferences, offering a flexible and responsive control mechanism. To reduce false readings caused by noise or temporary fluctuations, simple filtering methods such as averaging recent values are applied before decision-making [10-11].

The final processed data is displayed in real-time on a 16x2 LCD screen, which is interfaced with the Arduino using an I2C communication protocol. The screen refreshes every few seconds, presenting the user with updated values for soil moisture, nitrogen, phosphorus, and potassium. This localized display system eliminates the dependence on cloud connectivity, making it ideal for rural or remote farming areas where internet access may be limited or unreliable. In addition to visual output, the system can be configured to log data to local storage, such as an SD card module, for historical tracking and offline analysis [12] [14-17].

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This embedded data processing approach enables the system to function autonomously, intelligently interpreting sensor readings without the need for external computation or internet infrastructure. It also provides a scalable foundation for future enhancements, such as wireless data transmission or integration with mobile applications, while maintaining a simple and robust architecture suited to real-world agricultural environments.

C. Automatic Irrigation

The final phase of the system focuses on the automatic execution of irrigation based on real-time soil moisture data. This automation eliminates the need for manual watering routines, instead allowing the system to make intelligent irrigation decisions based on actual field conditions. Once the processed data from the capacitive soil moisture sensors indicate that the moisture level has fallen below a

At the initial pre-set threshold typically around 20–25% Volumetric Water Content (VWC), the Arduino microcontroller triggers the irrigation process. This decision-making logic is embedded within the microcontroller's firmware and continuously runs in the background, ensuring real-time responsiveness without requiring human intervention.

The system is designed to support multiple types of irrigation methods. The most prominent being drip irrigation and sprinkler irrigation. To overcome drip irrigation, water is delivered slowly and directly to the base of the plant through a network of emitters and pipes. This method is supposed to minimize evaporation and runoff, making it substantially effective for areas with water scarcity or crops with deep root systems. The automated system controls the mainline valve or pump that supplies pressure to the drip network, ensuring that water delivery is both timely and targeted. Fig. 6 is the graphical representation of automated irrigation system.



Fig. 6. Graphical Representation of Automates Irrigation [22]

In comparison, sprinkler irrigation involves spraying water over the crops in a manner like natural rainfall. This method is better suited for crops that require surface moisture or fields with irregular layouts. Sprinkler systems are effective for covering large areas and are often used in open-field farming for grains, vegetables, and turf grass. However, they are more susceptible to evaporation losses and can be less efficient than drip systems in hot or windy conditions. The system developed in this research can adapt to either method by switching the type of actuator and adjusting irrigation duration based on the delivery rate of each system.

To enhance flexibility, a manual override feature is included. This allows farmers to operate the system manually through a physical switch or optional user interface module, enabling control during emergencies or maintenance routines. Safety mechanisms are also integrated into the system, including time-based cut-offs to prevent the pump from running indefinitely in the event of sensor failure or dry-run protection to avoid damage to the pump in case of empty water tanks.

This automated irrigation approach, powered by sensor driven feedback and adaptable infrastructure, not only ensures that crops receive the right amount of water at the right time but also reduces labor and water waste. By integrating decision-making, real-time monitoring, and actuation in one self-contained system, this methodology represents a practical and scalable solution for smart agriculture. Whether deployed in small-scale farms or larger agricultural plots, the system enhances irrigation precision, boosts crop health, and contributes to more sustainable farming practices [12] [14].

VI. RESULTS AND DISCUSSION

The IoT-based soil monitoring and irrigation system demonstrated consistent performance in maintaining optimal soil conditions through real-time data acquisition and automated control. The soil moisture readings, as shown in Fig. 1, stayed largely within the optimal range of 25% to 40% during the observation period. The system automatically initiated irrigation when moisture levels dropped below the critical threshold of approximately 30%, resulting in a timely increase in soil moisture. This cycle confirms the reliability of the sensing and control mechanism, safeguarding adequate water release while avoiding over-irrigation.

In addition to moisture observation, the system efficiently traced soil productivity through nutrient-level monitoring devices. The distribution of nitrogen, phosphorus, and potassium concentrations is visualized in Fig. 2, as per available studies and data, with nitrogen ranging between 60–70 mg/kg, phosphorus between 20–30 mg/kg, and potassium between 160–180 mg/kg. These values persisted within adequate agricultural measures, indicating steady soil nutrition. Consistent nutrient levels over time suggest that the soil did not experience significant depletion, minimizing the need for frequent fertilization and supporting more sustainable farming practices.

By integrating both moisture and nutrient data, the system offers a comprehensive understanding of soil health. This dual-parameter monitoring enhances decision-making for irrigation and fertilization, promoting efficient resource usage and improved crop productivity. The results underscore the system's suitability for precision agriculture, particularly for small and medium-scale farms seeking affordable and reliable technological solutions.

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VII. CHALLENGES

There are some challenges while implementation and real time testing and analyzing the sensor data. Following are some challenges-

- Environmental factors affecting the performance like dust, waterlogging, and mechanical damage from farm equipment impacted on sensors.
- Internet and mobile network connectivity in remote areas is another challenge.
- Sensor calibration was difficult due to variations in soil type and sensor drift.
- The limited memory and processing power of the Arduino microcontroller constrained system scalability and the integration of advanced data analytics.
- Implementation of solar-powered energy systems to ensure sustainable, off-grid operation of the system.
- Consistent power supply in rural areas is another challenge.

IX. CONCLUSION AND FUTURE WORK

In this research author successfully demonstrated the design and implementation of an IoT-based soil monitoring and automatic irrigation system customized for agricultural purposes. By integration of soil moisture and nutrient sensors with an Arduino microcontroller, the system provided real-time monitoring of critical soil parameters and enabled efficient, automated irrigation based on actual field conditions.

The obtained results showed that the system maintained optimal soil moisture levels and offered consistent nutrient monitoring, contributing to water conservation and better soil fertility management. In the future the proposed system can enhance connectivity, analytics, and power management, the system holds strong potential for widespread adoption in diverse agricultural settings. The further extended work may be upgrading the microcontroller to a more powerful platform to support advanced data processing and machine learning capabilities. In addition, development of a mobile or web-based application for real-time remote monitoring and control of the irrigation system.

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